

Chambro

Room Architect

User Manual - v1.0.0

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Thanks for using **Chambro**.

Most reverbs describe a space in abstractions: size, density, a room dial from 0 to 100. Chambro asks for its **dimensions in metres**, and for what its walls are made of. Width, Length and Height are the literal size of the room being simulated, and they set the resonances the reverb is built from. Type in $9 \times 18 \times 8$ and the result is a chapel, because that's what a room that size does to sound.



Why a model of a room

I build instruments that model physical things, and one conviction runs through all of them: when the controls are physical quantities - a length, a material - the instrument stops being a parameter space and becomes a description. “Nine metres wide, stone walls” takes no hunting, because everyone has spent a lifetime inside rooms. Hearing carries a fast, unconscious model of how rooms behave, so a setting that matches a room that could exist sounds right, and the ear accepts it as a place instead of auditing it as an effect.

The usual way to put a real room into a mix is to record one: capture its impulse response and run the signal through it. That's a photograph - exact, and frozen. The room has one size, one material and one decay, all tied together the way they were on the day it was captured.

Chambro is a model instead, the way a 3D render is. It's less literal than a photograph, but every setting obeys the physics of rooms, so the result stays plausible. And because the settings are independent, plausible rooms that can't exist are within reach:

- the shape of a chapel with the walls of a tiled bathroom
- a stone crypt that rings for a full minute
- walls that are perfect mirrors, or perfect scatterers
- a room that grows from a vocal booth into a hall over the course of a song

How it works

At certain frequencies - a room's modes - sound bouncing between the walls lines up with itself and rings on. The dimensions alone set those frequencies, which is why a bathroom sings a note and why two rooms of the same volume but a different shape sound different.

Inside Chambro are sixteen delay lines, each tuned to one of those modes for the room on the dials. Move a wall and every line moves with it. The lines feed back into each other through a mixer, which turns sixteen ringing echoes into a smooth wash. The mixer is exactly lossless: it moves energy around without adding or removing any, so the only thing shortening the tail is the absorption that was asked for, and the Time dial means what it says.

The controls

Width, Length, Height

1 to 100 m each. They shape the character - colour, density, the way the room rings - more than the length of the tail. Shape matters as much as size: a cube is the worst-behaved room there is, because its modes pile onto the same few frequencies, while uneven proportions spread them out. A 30 × 30 × 2.8 m room is a car park, and it honks like one.

Clutter

How far the room is from an empty box. At 0% the lines sit exactly on the room's modes, which in a perfect shoebox stack up and ring metallic, with empty gaps between them. Turning it up spreads the lines evenly across the room's range: the stacks come apart and the gaps fill in, the way doors, beams and furniture break up a real room's standing waves. The default of 35% clears the metallic edge and keeps the character. Go higher for caves and irregular spaces, lower when the ring is the point.

Diffusion

How rough the walls are. Damping says how much they absorb; Diffusion says how much they scatter. At 0% every wall is a mirror and the sixteen lines ring on their own, like tuned combs. At 100% the walls scatter completely into a dense, even wash. In between, scattering arrives with frequency - the low end stays mirror-like while the top scatters - which is how a real surface behaves.

The tail gets shorter as Diffusion rises, because scattering moves energy onto faster-decaying modes. Below about 25% the result is an effect rather than a room, since real walls are rougher than that: smooth concrete sits around 50%, rough stone, pews and bookshelves towards 100%.

Time

The reverberation time, 0.1 to 60 s: how long the low end of the tail takes to fall by 60 dB. The top end is Damping's job. Time is most accurate towards the diffusive end; with Diffusion near 0% the room rings longer than the dial says.

Damping

How much treble the walls absorb, 0 to 100%. At 30% the 1 kHz tail lasts 70% as long as the bass. It's really a material dial: low is tile, glass and stone, high is carpet, curtains and people. With AUTO TIME on it shortens the whole decay as well, because soft materials soak up a lot of everything. A 4 × 5 × 3 m room, changing nothing but Damping:

Damping	material	AUTO TIME
12%	bare stone, tile	2.4 s
30%	plaster, bare wood	1.4 s
48%	a rug, some furniture	0.8 s

Predelay

The gap before the room answers, 0 to 250 ms, on the reverb only. A few milliseconds glue a source into a space; 20 to 40 ms keep a vocal's attack clear of the room; past 100 ms it reads as an echo.

Stereo

The width of the tail, 0 to 200%. 0% is exactly mono, 100% is the room's natural spread, and beyond that is wider than a real room. The dry signal stays put at every setting.

Dry, Wet

The two levels in dB, each muted at the bottom of its travel. Pull Dry all the way down on a send.

AUTO: let the room decide

With these two switches on, the room sets Time and Predelay by itself. They work independently, and turning a dial one of them drives switches that one off.

AUTO TIME uses Sabine's equation, the acousticians' standard for predicting how long a real room reverberates:

$$T60 = 0.161 \times \text{volume} \div (\text{surface} \times \text{absorption})$$

Volume and surface come from the room dimensions, absorption from Damping. Drag a wall out and the decay lengthens, as it would.

AUTO PREDELAY uses the mean free path - the average distance sound travels between two reflections, $4 \times \text{volume} \div \text{surface}$ - divided by the speed of sound. That's exactly the question predelay asks: how long before the room has answered at all. A vocal booth answers in 4 ms, a cathedral in 67.

Presets

Most of the sixteen presets are real kinds of space at real dimensions, with the decay Sabine's equation gives for their size and materials. That's why the Tiled Bathroom, at $2.2 \times 3.0 \times 2.5$ m, rings longer than the Drum Room at six times its volume: size sets the character, material sets the length. A few, like the Infinite Cavern and the Frozen Cathedral, are the impossible kind.

The AUTO switches are off in every preset. Switching them on gives the bare room, without the furniture and people the preset's own Time allows for. Step through the presets with the arrows, or click the name to browse.

Tips

- Start with both AUTO switches on and just build the room: dimensions for the size, Damping for the material. Switch them off to overrule the physics.
- Reach for Predelay before less Wet. Most "too much reverb" is really "reverb too early".
- Long tails settle onto pitches. That's what a real cavern does too.

Keyboard and screen readers

Tab and Shift-Tab move between controls, and the focused one draws a bright ring. Arrow keys nudge a dial (hold Shift for finer steps), Return on a dial opens a box to type a value, and Return or Space flips an AUTO switch. On the preset bar the arrows step through the presets and Return or Space opens the list.

Chambro works with VoiceOver on macOS and with NVDA, JAWS and Narrator on Windows. Every control has a spoken name and value, and each AUTO switch carries a name of its own, apart from the dial it drives.

Demo and license

There's one download, and it's the full plugin. Until a license is installed the output fades to silence for two seconds every 45 seconds; nothing else is held back.

A purchase comes with a license file by email. Click **DEMO - INSTALL LICENSE** at the foot of the window and pick the file: the fades stop, and the strip reads **LICENSED TO** and the buyer's name. The file is checked on the computer itself, with no internet connection, account or activation, and it keeps working on a new machine. Keep a copy somewhere safe.

Licenses are stored in ~/Library/Application Support/StefanoBaldan/Licenses on macOS and in %LOCALAPPDATA%\StefanoBaldan\Licenses on Windows.

Compatibility

Formats: VST3, Audio Unit (AU), CLAP

macOS: 10.15 or later (Apple Silicon and Intel)

Windows: 64-bit

If the DAW doesn't list Chambro right after installing, rescan the plug-ins.

Have fun building rooms.

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